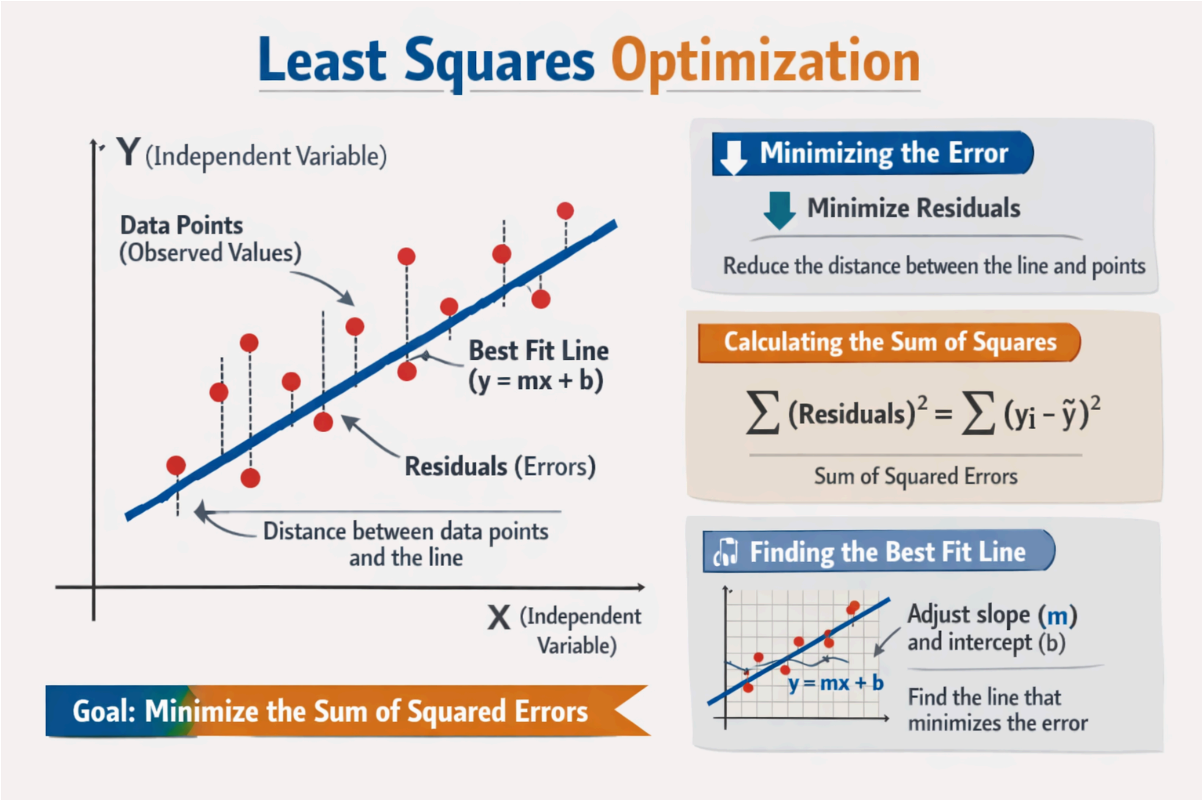
Least Squares Optimization



The key idea is simple:

- Each red point represents observed data.
- The blue line is the regression line we're trying to fit.
- The dashed vertical lines are the residuals (differences between observed and predicted values).
- Least squares optimization adjusts the slope and intercept of the line so that the sum of squared residuals is as small as possible.

This process ensures the line is the “best fit” in terms of minimizing overall error.

Example 1: Linear Regression (Best Fit Line)

Given points: $(1,1),(-2,-1),(3,2)$.

1. **Calculate the means of x and y:**
   1. $\bar{x} = \frac{1+(-2)+3}{3} = \frac{2}{3}$
   2. $\bar{y} = \frac{1+(-1)+2}{3} = \frac{2}{3}$
2. **Calculate the slope (b or m):**
   1. $m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$
   2. Numerator: $(1-\frac{2}{3})(1-\frac{2}{3}) + (-2-\frac{2}{3})(-1-\frac{2}{3}) + (3-\frac{2}{3})(2-\frac{2}{3}) = \frac{23}{3}$
   3. Denominator: $(1-\frac{2}{3})^2 + (-2-\frac{2}{3})^2 + (3-\frac{2}{3})^2 = \frac{38}{3}$
   4. $m = \frac{23/3}{38/3} = \frac{23}{38}$
3. **Calculate the intercept (a or c):**
   1. $c = \bar{y} - m\bar{x} = \frac{2}{3} - \frac{23}{38} \times \frac{2}{3} = \frac{5}{19}$
4. **Equation of the line:** $y = \frac{23}{38}x + \frac{5}{19}$.

Example 2: Data Modeling (Height/Weight)

Find the best-fit line for: Heights $x = [160, 162, 164, 166, 168]$, Weights $y = [52, 55, 57, 60, 61]$

1. **Mean values:** $\bar{x} = 164$, $\bar{y} = 57$.
2. **Calculations:**
    1. $\sum (X - x_i)(Y - y_i) = 46$
    2. $\sum (X - x_i)^2 = 40$
3. **Slope ($\boldsymbol{m}$):** $m = \dfrac{46}{40} = 1.15$
4. **Intercept ($\boldsymbol{c}$):** $c = \bar{y} - m\bar{x} = 57 - 1.15 \times 164 = -131.6$
5. **Equation:** $y = 1.15x - 131.6$.

Problem Setup

We want to fit a straight line:

$$y = \beta_0 + \beta_1 x$$

to a given dataset (x_i, y_i) using least squares.

Matrix Formulation

Define:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

The model is:

$$y = X\beta + \epsilon$$

Least squares solution:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Normal Equations Form

Expanding:

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Solve this 2×2 system for β_0, β_1.

Worked Example

Suppose data points: $(1,2), (2,3), (3,5)$.

Step 1: Construct matrices

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

Step 2: Compute normal equations

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

So:

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

Step 3: Solve system

Determinant = $3 \cdot 14 - 6 \cdot 6 = 6$.

Inverse:

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

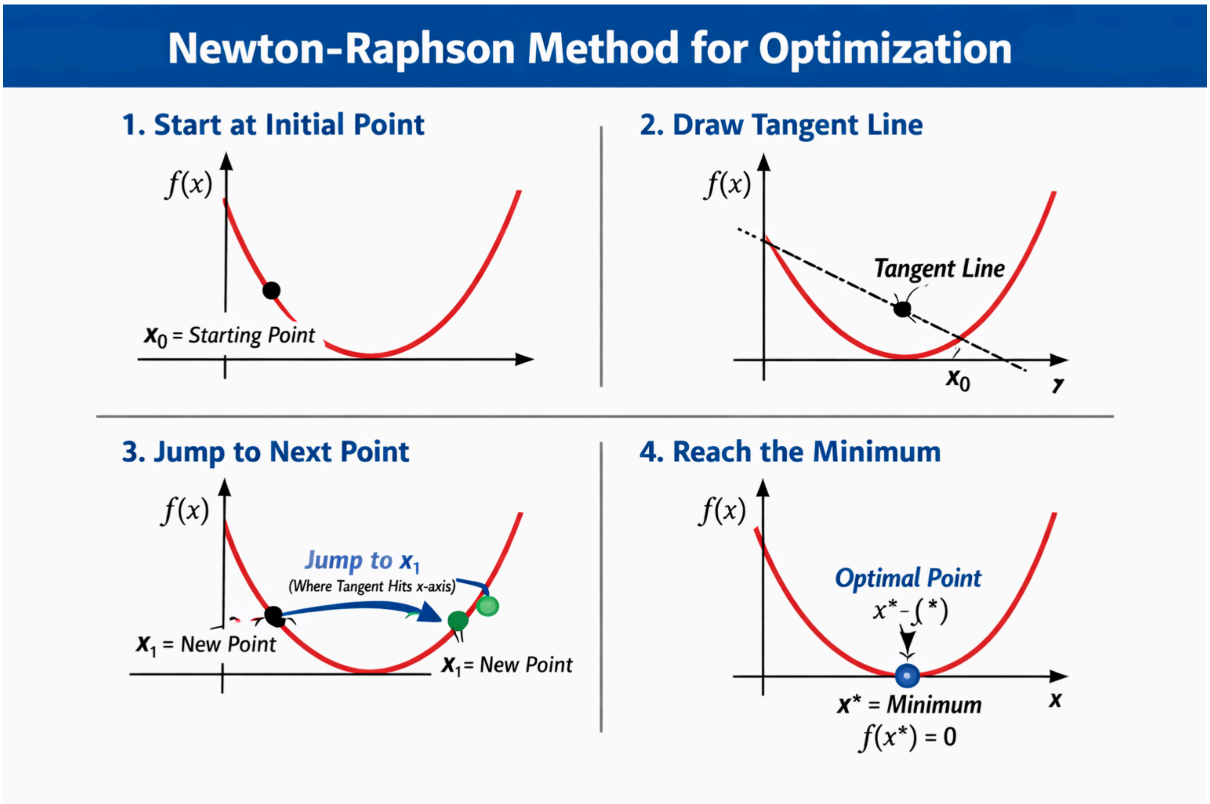
Multiply:

$$\hat{\beta} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 23 \end{bmatrix} = \begin{bmatrix} 0.333 \\ 1.5 \end{bmatrix}$$

◆ Final Fitted Line

$$y \approx 0.333 + 1.5x$$

Unconstrained Optimization Technique:



1. **Start at an initial point**
   - Pick a starting guess x_0 on the curve $f(x)$.
   - This is where you begin your search for the minimum.
2. **Draw the tangent line**
   - At x_0, draw the tangent to the curve.
   - The slope of this tangent is $f'(x_0)$.
   - The tangent line represents the local linear approximation of the curve.
3. **Jump to the next point**
   - Extend the tangent line until it meets the x-axis.
   - The intersection point becomes your next estimate x_1.
   - This step uses the formula:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

4. **Reach the minimum**
   - Repeat the process until the tangent becomes flat (slope ≈ 0).
   - The point where the tangent is horizontal is the **optimal point** x^*.
   - At this point, $f'(x^*) = 0$ and $f''(x^*) > 0$, confirming a minimum.

Refer Link : Pg. 2.22 onwards for numericals

2.4 NEWTON—RAPHSON METHOD

Let $f(x) = 0$ be the given equation and x_0 be an approximate root of the equation. If $x_1 = x_0 + h$ be the exact root then $f(x_1) = 0$.

i.e., $$f(x_0 + h) = 0$$

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \cdots = 0 \qquad \text{[By Taylor's series]}$$

Since h is small, neglecting h^2 and higher powers of h,

$$f(x_0) + h\, f'(x_0) = 0$$

$$h = -\frac{f(x_0)}{f'(x_0)}$$

$$\therefore \qquad x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Similarly, starting with x_1, a still better approximation x_2 is obtained.

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

In general, $$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This equation is known as the *Newton–Raphson formula* or *Newton's iteration formula*.

2.4.1 Geometrical Interpretation

Let x_0 be a point near the root α of the equation $f(x) = 0$ (Fig. 2.3). The equation of the tangent at $P_0[x_0, f(x_0)]$ is

$$y - f(x_0) = f'(x_0)(x - x_0)$$

This line cuts the x-axis at x_1.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

which is a first approximation to the root α.

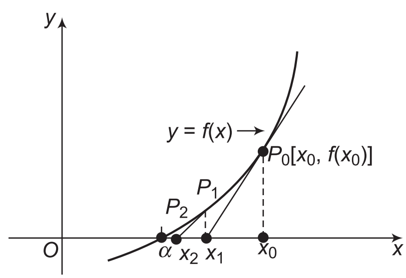


Fig. 2.3

If P_1 is the point corresponding to x_1 on the curve then the tangent at P_1 will cut the x-axis at x_2 which is nearer to α and is the second approximation to the root. Repeating this process, the root α is approached quite rapidly. Thus, this method consists of replacing the part of the curve between the point P_0 and the x-axis by means of the tangent to the curve at P_0.

2.4.2 Convergence of the Newton—Raphson Method

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = \phi(x_n) \qquad \text{...(2.1)}$$

The Newton–Raphson method converges if $|\phi'(x)| < 1$.

$$\phi(x) = x - \frac{f(x)}{f'(x)}$$

$$\phi'(x) = 1 - \left[\frac{[f'(x)]^2 - f(x)\, f''(x)}{[f'(x)]^2}\right] = \frac{f(x)\, f''(x)}{[f'(x)]^2}$$

$$|\phi'(x)| = \left|\frac{f(x)\, f''(x)}{[f'(x)]^2}\right|$$

Hence, the Newton–Raphson method converges if

$$\left|\frac{f(x)\, f''(x)}{[f'(x)]^2}\right| < 1$$

$$|f(x)\, f''(x)| < [f'(x)]^2 \qquad \text{...(2.2)}$$

If α is the actual root of $f(x) = 0$, a small interval should be selected in which $f(x)$, $f'(x)$ and $f''(x)$ are all continuous and the condition given by Eq. (2.2) is satisfied.

Hence, the Newton–Raphson method always converges provided the initial approximation x_0 is taken very close to the actual root α.

2.4.3 Rate of Convergence of the Newton—Raphson Method

Let α be exact root of $f(x) = 0$ and let x_n, x_{n+1} be two successive approximations to the actual root. If $\in_n$ and $\in_{n+1}$ are the corresponding errors then

$$x_n = \alpha + \in_n$$

$$x_{n+1} = \alpha + \in_{n+1}$$

Substituting in Eq. (2.1),

$$\alpha + \in_{n+1} = \alpha + \in_n - \frac{f(\alpha + \in_n)}{f'(\alpha + \in_n)}$$

$$\in_{n+1} = \in_n - \frac{f(\alpha + \in_n)}{f'(\alpha + \in_n)}$$

$$= \in_n - \frac{f(\alpha) + \in_n f'(\alpha) + \frac{\in_n^2}{2!} f''(\alpha) + \cdots}{f'(\alpha) + \in_n f''(\alpha) + \cdots} \qquad \text{[By Taylor's series]}$$

$$= \in_n - \frac{\in_n f'(\alpha) + \frac{\in_n^2}{2} f''(\alpha) + \cdots}{f'(\alpha) + \in_n f''(\alpha)} \qquad [\because \ f(\alpha) = 0]$$

Neglecting the derivatives of order higher than two,

$$\begin{aligned}
\epsilon_{n+1} &= \epsilon_n - \frac{\epsilon_n f'(\alpha) + \dfrac{\epsilon_n^2}{2} f''(\alpha)}{f'(\alpha) + \epsilon_n f''(\alpha)} \\
&= \frac{1}{2}\left[\frac{\epsilon_n^2 f''(\alpha)}{f'(\alpha) + \epsilon_n f''(\alpha)}\right] \\
&= \frac{\epsilon_n^2}{2}\left[\frac{\dfrac{f''(\alpha)}{f'(\alpha)}}{1 + \epsilon_n \dfrac{f''(\alpha)}{f'(\alpha)}}\right] \\
&\approx \frac{\epsilon_n^2}{2} \frac{f''(\alpha)}{f'(\alpha)} \qquad \text{...(2.3)}
\end{aligned}$$

Equation (2.3) shows that the error at each stage is proportional to the square of the error in the previous stage. Hence, the Newton–Raphson method has a quadratic convergence and the convergence is of the order 2.

Example 1

Find the root of the equation $x^3 + x - 1 = 0$, *correct up to four decimal places.*

Solution

Let $f(x) = x^3 + x - 1$

$f(0) = -1$ and $f(1) = 1$

Since $f(0) < 0$ and $f(1) > 0$, the root lies between 0 and 1.

Let $x_0 = 1$

$$f'(x) = 3x^2 + 1$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_0) = f(1) = 1$$

$$f'(x_0) = f'(1) = 4$$

$$\begin{aligned}
x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
&= 1 - \frac{1}{4} \\
&= 0.75
\end{aligned}$$

$$f(x_1) = f(0.75) = 0.171875$$
$$f'(x_1) = f'(0.75) = 2.6875$$
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$
$$= 0.75 - \frac{0.171875}{2.6875}$$
$$= 0.68605$$
$$f(x_2) = f(0.68605) = 0.00894$$
$$f'(x_2) = f'(0.68605) = 2.41198$$
$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$
$$= 0.68605 - \frac{0.00894}{2.41198}$$
$$= 0.68234$$
$$f(x_3) = f(0.68234) = 0.000028$$
$$f'(x_3) = f'(0.68234) = 2.39676$$
$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$
$$= 0.68234 - \frac{0.000028}{2.39676}$$
$$= 0.68233$$

Since x_3 and x_4 are same up to four decimal places, the root is 0.6823.

Example 2

Find a root of $x^4 - x^3 + 10x + 7 = 0$, correct up to three decimal places between –2 and –1 by the Newton–Raphson method.

Solution

Let $\quad f(x) = x^4 - x^3 + 10x + 7$

The root lies between –2 and –1.

Let $x_0 = -2$

$$f'(x) = 4x^3 - 3x^2 + 10$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$
$$f(x_0) = f(-2) = 11$$

$$f'(x_0) = f'(-2) = -34$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= -2 - \frac{11}{(-34)}$$

$$= -1.6765$$

$$f(x_1) = f(-1.6765) = 2.8468$$

$$f'(x_1) = f'(-1.6765) = -17.2802$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= -1.6765 - \frac{2.8468}{(-17.2802)}$$

$$= -1.5118$$

$$f(x_2) = f(-1.5118) = 0.561$$

$$f'(x_2) = f'(-1.5118) = -10.6777$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= -1.5118 - \frac{0.561}{(-10.6777)}$$

$$= -1.4593$$

$$f(x_3) = f(-1.4593) = 0.0497$$

$$f'(x_3) = f(-1.4593) = -8.8193$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$= -1.4593 - \frac{0.0497}{(-8.8193)}$$

$$= -1.4537$$

$$f(x_4) = f(-1.4537) = 0.0008$$

$$f'(x_4) = f'(-1.4537) = -8.6278$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)}$$

$$= -1.4537 - \frac{0.0008}{(-8.6278)}$$

$$= -1.4536$$

Since x_4 and x_5 are same up to three decimal places, a root is –1.453.

Example 3

Find the root of $x^4 - x - 10 = 0$, *correct up to three decimal places.*

Solution

Let $$f(x) = x^4 - x - 10$$

$$f(1) = -10, \quad \text{and} \quad f(2) = 4$$

Since $f(1) < 0$ and $f(2) > 0$, the root lies between 1 and 2.

Let $x_0 = 2$

$$f'(x) = 4x^3 - 1$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_0) = f(2) = 4$$

$$f'(x_0) = f'(2) = 31$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 2 - \frac{4}{31}$$

$$= 1.871$$

$$f(x_1) = f(1.871) = 0.3835$$

$$f'(x_1) = f'(1.871) = 25.1988$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.871 - \frac{0.3835}{25.1988}$$

$$= 1.8558$$

$$f(x_2) = f(1.8558) = 5.2922 \times 10^{-3}$$

$$f'(x_2) = f'(1.8558) = 24.5655$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 1.8558 - \frac{5.2922 \times 10^{-3}}{24.5655}$$

$$= 1.8556$$

Since x_2 and x_3 are same up to three decimal places, the root is 1.855.

Example 4

Find the real root of $x \log_{10} x - 1.2 = 0$, *correct up to three decimal places.* **[Summer 2015]**

Solution

Let $$f(x) = x \log_{10} x - 1.2$$

$$f(1) = -1.2,\ f(2) = -0.5979 \quad \text{and} \quad f(3) = 0.2314$$

Since $f(2) < 0$ and $f(3) > 0$, the root lies between 2 and 3.

Let $$x_0 = 3$$

$$f'(x) = \log_{10} x + x \frac{1}{x \log_e 10} = \log_{10} x + \log_{10} e = \log_{10} x + 0.4343$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_0) = f(3) = 0.2314$$

$$f'(x_0) = f'(3) = 0.9114$$

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 3 - \frac{0.2314}{0.9114} \\ &= 2.7461 \end{aligned}$$

$$f(x_1) = f(2.7461) = 4.759 \times 10^{-3}$$

$$f'(x_1) = f'(2.7461) = 0.8730$$

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 2.7461 - \frac{4.759 \times 10^{-3}}{0.8730} \\ &= 2.7406 \end{aligned}$$

$$f(x_2) = f(2.7406) = -4.0202 \times 10^{-5}$$

$$f'(x_2) = f'(2.7406) = 0.8721$$

$$\begin{aligned} x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\ &= 2.7406 - \frac{(-4.0202 \times 10^{-5})}{0.8721} \\ &= 2.7406 \end{aligned}$$

Since x_2 and x_3 are the same up to three decimal places, the real root is 2.7406.

Example 5

Find a root between 0 *and* 1 *of the equation* $e^x \sin x = 1$, *correct up to four decimal places.*

Solution

Let $$f(x) = e^x \sin x - 1$$

$$f(0) = -1 \quad \text{and} \quad f(1) = 1.28$$

Since $f(0) < 0$ and $f(1) > 0$, the root lies between 0 and 1.

Let $x_0 = 0$

$$f'(x) = e^x (\cos x + \sin x)$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_0) = f(0) = -1$$

$$f'(x_0) = f'(0) = 1$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0 - \frac{(-1)}{1}$$

$$= 1$$

$$f(x_1) = f(1) = 1.2874$$

$$f'(x_1) = f'(1) = 3.7560$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1 - \frac{1.2874}{3.7560}$$

$$= 0.6572$$

$$f(x_2) = f(0.6572) = 0.1787$$

$$f'(x_2) = f'(0.6572) = 2.7062$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.6572 - \frac{0.1787}{2.7062}$$

$$= 0.5912$$

$$f(x_3) = f(0.5912) = 6.6742 \times 10^{-3}$$
$$f'(x_3) = f'(0.5912) = 2.5063$$
$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$
$$= 0.5912 - \frac{6.6742 \times 10^{-3}}{2.5063}$$
$$= 0.5885$$
$$f(x_4) = f(0.5885) = -8.1802 \times 10^{-5}$$
$$f'(x_4) = f'(0.5885) = 2.4982$$
$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)}$$
$$= 0.5885 - \frac{-8.1802 \times 10^{-5}}{2.4982}$$
$$= 0.5885$$

Since x_4 and x_5 are the same up to four decimal places, the root is 0.5885.

Example 6

Find the real root of the equation $3x = \cos x + 1$, *correct up to four decimal places.*

Solution

Let $\quad f(x) = 3x - \cos x - 1$

$$f(0) = -2 \quad \text{and} \quad f(1) = 1.4597$$

Since $f(0) < 0$ and $f(1) > 0$, the root lies between 0 and 1.

Let $x_0 = 1$

$$f'(x) = 3 + \sin x$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$
$$f(x_0) = f(1) = 1.4597$$
$$f'(x_0) = f'(1) = 3.8415$$
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$
$$= 1 - \frac{1.4597}{3.8415}$$
$$= 0.62$$

$$f(x_1) = f(0.62) = 0.0461$$
$$f'(x_1) = f'(0.62) = 3.5810$$
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$
$$= 0.62 - \frac{0.0461}{3.5810}$$
$$= 0.6071$$
$$f(x_2) = f(0.6071) = -5.8845 \times 10^{-6}$$
$$f'(x_2) = f'(0.6071) = 3.5705$$
$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$
$$= 0.6071 - \frac{-5.8845 \times 10^{-6}}{3.5705}$$
$$= 0.6071$$

Since x_2 and x_3 are the same up to four decimal places, the real root is 0.6071.

Example 7

Find the real positive root of the equation $x \sin x + \cos x = 0$, *which is near* $x = \pi$ *correct up to four significant digits.* **[Summer 2015]**

Solution

Let $f(x) = x \sin x + \cos x$

Let $x_0 = \pi$

$$f'(x) = x \cos x + \sin x - \sin x = x \cos x$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$
$$f(x_0) = f(\pi) = -1$$
$$f'(x_0) = f'(\pi) = -\pi$$
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$
$$= \pi - \frac{(-1)}{(-\pi)}$$
$$= 2.82328$$

$$f(x_1) = f(2.82328) = -0.06618$$
$$f'(x_1) = f'(2.823287) = -2.68145$$
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$
$$= 2.82328 - \frac{(-0.06618)}{(-2.68145)}$$
$$= 2.7986$$
$$f(x_2) = f(2.7986) = -0.00056$$
$$f'(x_2) = f'(2.7986) = -2.63559$$
$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$
$$= 2.7986 - \frac{(-0.00056)}{(-2.63559)}$$
$$= 2.79839$$
$$f(x_3) = f(2.79839) = -0.0001$$
$$f'(x_3) = f'(2.79839) = -2.63519$$
$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$
$$= 2.79839 - \frac{(-0.0001)}{(-2.63519)}$$
$$= 2.79839$$

Since x_3 and x_4 are same up to four decimal point, the root is 2.7983.

Example 8

Find the positive root of $x = \cos x$ *using Newton's method correct to three decimal places.*

Solution

Let $\qquad f(x) = x - \cos x$

$$f(0) = -1 \quad \text{and} \quad f(1) = 0.4597$$

Since $f(0) < 0$ and $f(1) > 0$, the root lies between 0 and 1.

Let $\qquad x_0 = 1$

$$f'(x) = 1 + \sin x$$

By the Newton–Raphson method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_0) = f(1) = 0.4597$$

$$f'(x_0) = f'(1) = 1.8415$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1 - \frac{0.4597}{1.8415}$$

$$= 0.7504$$

$$f(x_1) = f(0.7504) = 0.019$$

$$f'(x_1) = f'(0.7504) = 1.6819$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.7504 - \frac{0.019}{1.6819}$$

$$= 0.7391$$

$$f(x_2) = f(0.7391) = 0.00002$$

$$f'(x_2) = f'(0.7391) = 1.6736$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.7391 - \frac{0.00002}{1.6736}$$

$$= 0.7391$$

Since x_2 and x_3 are same up to three decimal places, the root is 0.739.

Example 9

Derive the iteration formula for $\sqrt{N}$ *and, hence, find*

(i) $\sqrt{28}$ **[Summer 2015]**

(ii) $\sqrt{65}$ **[Winter 2014]**

(iii) $\sqrt{3}$ **[Winter 2014]**

correct up to three decimal places.

Solution

Let $$x = \sqrt{N}$$

$$x^2 - N = 0$$

Let $$f(x) = x^2 - N$$

$$f'(x) = 2x$$

By the Newton–Raphson method,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n^2 - N}{2x_n} \\ &= \frac{x_n^2 + N}{2x_n} \end{aligned}$$

This is the iteration formula for $\sqrt{N}$.

(i) For $N = 28$, $f(x) = x^2 - 28$

$$f(5) = -3 \quad \text{and} \quad f(6) = 8$$

Since $f(5) < 0$ and $f(6) > 0$, the root lies between 5 and 6.

Let $x_0 = 5$

$$\begin{aligned} x_{n+1} &= \frac{x_n^2 + 28}{2x_n} \\ x_1 &= \frac{x_0^2 + 28}{2x_0} = 5.3 \\ x_2 &= \frac{x_1^2 + 28}{2x_1} = 5.2915 \\ x_3 &= \frac{x_2^2 + 28}{2x_2} = 5.2915 \end{aligned}$$

Since x_2 and x_3 are same up to three decimal places,

$$\sqrt{28} = 5.2915$$

(ii) For $N = 65, f(x) = x^2 - 65$

$$f(8) = -1 \quad \text{and} \quad f(9) = 16$$

Since $f(8) < 0$ and $f(9) > 0$, the root lies between 8 and 9.

Let $x_0 = 8$

$$x_{n+1} = \frac{x_n^2 + 65}{2x_n}$$

$$x_1 = \frac{x_0^2 + 65}{2x_0} = 8.0625$$

$$x_2 = \frac{x_1^2 + 65}{2x_1} = 8.0623$$

Since x_1 and x_2 are same up to three decimal places,

$$\sqrt{65} = 8.0623$$

(iii) For $N = \sqrt{3},\ f(x) = x^2 - 3$

$$f(1) = -2 \quad \text{and} \quad f(2) = 1$$

Since $f(1) < 0$ and $f(2) > 0$, the root lies between 1 and 2.

Let $x_0 = 2$

$$x_{n+1} = \frac{x_n^2 + 3}{2x_n}$$

$$x_1 = \frac{x_0^2 + 3}{2x_0} = 1.75$$

$$x_2 = \frac{x_1^2 + 3}{2x_1} = 1.7321$$

$$x_3 = \frac{x_2^2 + 3}{2x_2} = 1.7321$$

Since x_2 and x_3 are same up to three decimal places,

$$\sqrt{3} = 1.7321$$

Example 10

Find an iterative formula for $\sqrt[k]{N}$ *, where N is a positive number and hence, evaluate (i)* $\sqrt[3]{11}$*, and (ii)* $\sqrt[3]{58}$ **[Summer 2015]**

Solution

Let $x = \sqrt[k]{N}$

$$x^k - N = 0$$

Let $x_0 = 4$

$$x_{n+1} = \frac{2x_n^3 + 58}{3x_n^2}$$

$$x_1 = \frac{2x_0^3 + 58}{3x_0^2} = 3.875$$

$$x_2 = \frac{2x_1^3 + 58}{3x_1^2} = 3.8709$$

$$x_3 = \frac{2x_2^3 + 58}{3x_2^2} = 3.8709$$

Since x_2 and x_3 are same up to four decimal places,

$$\sqrt[3]{58} = 3.8709$$

EXERCISE 2.3

I. Find the roots of the following equations using the Newton–Raphson method:

1. $x^3 - x - 1 = 0$

[Ans.: 1.3247]

2. $x^3 + 2x^2 + 50x + 7 = 0$

[Ans.: –0.1407]

3. $x^3 - 5x + 3 = 0$

[Ans.: 0.6566]

4. $x^4 - x - 9 = 0$

[Ans.: 1.8134]

5. $\cos x - xe^x = 0$

[Ans.: 0.5177]

6. $x \log_{10} x = 4.772393$

[Ans.: 6.0851]

7. $x - 2\sin x = 0$

[Ans.: 1.8955]

8. $x \tan x = 1.28$

[Ans.: 6.4783]

9. $\cos x = x^2$

[Ans.: 0.8241]